

ANNAMALAI AJAY.D

 : PORTFOLIO

 : GITHUB

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PROFILE SUMMARY

Aspiring Machine Learning Developer / Software Developer Engineering Intern with strong skills in Python, machine learning, and software development. Hands-on experience in data collection, data preprocessing, model training, model evaluation, predictive analytics, and deployment. Familiar with the complete software development lifecycle, backend integration, unit testing, and Agile methodologies. Highly motivated to contribute to production-ready AI-driven systems and scalable applications

TECHNICAL SKILLS

Programming Languages: Python (Proficient), Java (Basic), C++ (Basic), JavaScript (Basic)

Machine Learning & AI: Supervised and Unsupervised Learning, CNN, RNN, LSTM

ML Libraries: Scikit-learn, TensorFlow, PyTorch (Basic)

Data Engineering & Analysis: Data preprocessing, feature engineering, EDA, Pandas, NumPy, Matplotlib

Performance Metrics: Accuracy, Precision, Recall, F1-score

Software Development: Clean code practices, Unit Testing, Integration Testing, Debugging

Web & Backend: React (Basic), Node.js (Basic), Flask, Django

Databases: MySQL, PostgreSQL, MongoDB

Version Control Systems: Git, GitHub

Cloud Platforms: AWS, GCP, Azure (Fundamentals)

CORE COMPETENCIES

- Collaboration and teamwork in cross-functional environments
- Strong problem-solving skills and analytical thinking
- Attention to detail in data preprocessing, model evaluation, and testing
- Highly motivated with a continuous learning mindset

WORK EXPERIENCE

IoT-Based Real-Time Health Monitoring System (Aug 2024 – Nov 2024)

Developed machine learning models using Python and Scikit-learn for predictive analytics on real-time sensor data. Collected and analyzed large datasets, performed data preprocessing and normalization, and applied machine learning techniques for prediction. Integrated models into a production-like pipeline and visualized outputs through a React-based dashboard. Evaluated models using performance metrics and documented results.

Edge-AI Visual Inspection System (May 2024 – Jul 2024)

Implemented CNN-based deep learning models using TensorFlow to automate defect detection in electronic components. Conducted model training, performance evaluation, and optimization to improve accuracy and efficiency. Visualized results using Matplotlib and maintained technical documentation for model performance.

Smart Inventory Management System (Jan 2024 – Apr 2024)

Worked through the complete software development lifecycle to build a reliable backend system. Implemented database solutions using MySQL and MongoDB, developed CRUD operations, performed unit and integration testing, and managed version control using Git and GitHub following Agile practices.

EDUCATION

Bachelor of Engineering (B.E.) – Electronics and Communication Engineering Sathyabama Institute of Science and Technology, Chennai (2023 – 2027) CGPA: 8.01

Higher Secondary Certificate (HSC – Class XII) – DEVA ANNAI Matriculation Higher Secondary School, Madurai (2022 – 2023) Percentage: 70%

AWARD AND CERTIFICATION

Best Project Showcase Award for the IoT-Based Real-Time Health Monitoring System, recognized for effective integration of machine learning models, real-time data preprocessing, and visualization.

Tech Ideathon Finalist for proposing a machine learning-driven predictive analytics solution, highlighting innovation, applied AI skills, and technical creativity.

Project Excellence Recognition for implementing end-to-end ML workflows, including data collection, model training, evaluation, and deployment concepts.

Technical Merit Recognition for applying clean coding practices, unit testing, and version control (Git/GitHub) in software development projects.

DECLARATION

I hereby declare that the above information is true and correct to the best of my knowledge